

September 2026 Exam

EC400

Probability and Statistical Inference

Suitable for all candidates

Instructions to candidates

This paper contains **SIX** questions. Answer **ANY FOUR**.

Each question carries 25 marks; the four attempted questions carry 100 marks in total.

Show enough reasoning and working to justify your answers.

If more than four questions are attempted, the four answers awarded the highest marks will count toward the examination mark.

Time Allowed	Reading Time: <i>None</i>
	Writing Time: <i>2 hours</i>

You are supplied with:	<i>No additional materials</i>
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You may also use:	<i>No additional materials</i>
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Calculators:	<i>Calculators are not allowed in this examination</i>
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If you feel that you need to make an additional assumption, state it clearly before proceeding.

Question 1

Let W describe whether a computer system is working: $W = 1$ if it is working and $W = 0$ if it is not. Let $P(W = 0) = \pi$, where $0 < \pi < 1$. There are two monitoring devices. Let A be the event that the first device sounds its alarm, and let B be the event that the second device sounds its alarm. Write

$$P(A | W = 1) = a_1, \quad P(A | W = 0) = a_0, \quad P(B | W = 1) = b_1, \quad P(B | W = 0) = b_0,$$

where $0 < a_1 < a_0 < 1$ and $0 < b_1 < b_0 < 1$. Assume that the alarms are independent conditional on $W = 0$, and also independent conditional on $W = 1$. For an observed event C for which the following ratio is defined, let

$$\text{odds that the system does not work after observing } C := \frac{P(W = 0 | C)}{P(W = 1 | C)}.$$

[25 marks]

- (a) Write each of the following in terms of π , a_0 and a_1 : $P(A)$, $P(W = 0 | A)$, and the odds that the system does not work after observing A . **[5 marks]**
- (b) Show that observing A increases the probability that the system is not working. Explain how the assumption $a_0 > a_1$ leads to this result. **[4 marks]**
- (c) Derive the odds that the system does not work after observing both A and B . Identify exactly where conditional independence is used. **[5 marks]**
- (d) Let I_A and I_B be the indicator (dummy) variables for the events A and B : $I_A = 1$ if A occurs and $I_A = 0$ otherwise, with I_B defined similarly. Prove that

$$\text{Cov}(I_A, I_B) = \pi(1 - \pi)(a_0 - a_1)(b_0 - b_1).$$

Are the alarms independent (without conditioning on W)? Give some intuition for why or why not, noting that they are independent conditional on W . *Hint*: note that $\mathbb{E}[I_A] = 1 \times P(A) + 0 \times P(A^c) = P(A)$. As such, start by writing $\mathbb{E}[I_A I_B]$ as a probability. **[6 marks]**

- (e) Let B^c denote the event that the second alarm does not sound. Derive $P(W = 0 | A \cap B^c)$ and the necessary and sufficient condition under which this probability is greater than its prior value π , and prove that it is less than $P(W = 0 | A)$. Explain the two comparisons in words. **[5 marks]**

Question 2

A continuous random variable X has density

$$f_X(x) = 3x^2, \quad 0 < x < 1,$$

and density zero otherwise. Define the odds transformation

$$Y = \frac{X}{1 - X}.$$

[25 marks]

- (a) Verify that f_X is a density and derive the cumulative distribution function (CDF) of X . **[4 marks]**
- (b) State the possible values of Y (its support). Show that the probability density of Y is given by
- $$f_Y(y) = \frac{3y^2}{(1+y)^4}.$$
- [6 marks]**
- (c) For $q \in (0, 1)$, derive the q th quantile of Y . State the median as a special case. **[4 marks]**
- (d) Use the CDF of Y to calculate $P(Y > 1)$ and $P(1 < Y < 3)$. **[5 marks]**
- (e) Calculate the values of $r > 0$ for which $\mathbb{E}[Y^r]$ is finite.
Hint: First write the expectation as an integral from 0 to 1, plus an integral from 1 to ∞ . For $y \geq 1$, use $y \leq 1 + y \leq 2y$ to compare the integrand with y^{r-2} . You may use the fact that $\int_1^\infty y^p dy$ is finite if and only if $p < -1$. **[6 marks]**

Question 3

Let (X, Y) be distributed with joint density equal to

$$f_{X,Y}(x, y) = e^{-y}, \quad 0 < x < y < \infty,$$

and equal to zero otherwise.

[25 marks]

- (a) Show that the density is non-negative and that its integral over the support is equal to one. What are the marginal densities $f_X(x)$ and $f_Y(y)$? **[6 marks]**
- (b) Derive $f_{X|Y}(x | y)$. What standard distribution does $X | Y = y$ have? Calculate $\mathbb{E}[X | Y = y]$ and $\text{Var}(X | Y = y)$. **[5 marks]**
- (c) From part (a), $Y \sim \text{Gamma}(2, 1)$, with $\mathbb{E}[Y] = 2$ and $\text{Var}(Y) = 2$. Use the law of iterated expectations and the law of total variance, together with the conditional moments in part (b), to calculate $\mathbb{E}[X]$ and $\text{Var}(X)$. **[4 marks]**
- (d) Calculate $\text{Cov}(X, Y)$. Are X and Y independent? Justify your answer. **[4 marks]**
- (e) Derive $f_{Y|X}(y | x)$. Define the gap $G = Y - X$ and use this conditional density to find the distribution of $G | X = x$. Hence determine whether X and G are independent. **[6 marks]**

Question 4

Consider the linear regression model

$$y = X\beta + \varepsilon, \quad \mathbb{E}[\varepsilon | X] = 0, \quad \hat{\beta} = (X'X)^{-1}X'y,$$

where X has full column rank. The observations are independent conditional on X , but their error variances may differ:

$$\Omega = \text{Var}(\varepsilon | X) = \text{diag}(\sigma_1^2, \dots, \sigma_n^2).$$

Here $\text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ denotes the $n \times n$ diagonal matrix with diagonal entries $\sigma_1^2, \dots, \sigma_n^2$. An economist wants to test a scalar restriction

$$H_0 : a'\beta = \theta_0 \quad \text{against} \quad H_1 : a'\beta \neq \theta_0, \quad \hat{\theta} = a'\hat{\beta},$$

where a is a fixed non-zero vector.

[25 marks]

- (a) Show that $\mathbb{E}[\hat{\beta} \mid X] = \beta$. Make sure to point out where you are using any assumptions. **[6 marks]**
- (b) Derive
- $$\text{Var}(\hat{\beta} \mid X) = (X'X)^{-1}X'\Omega X(X'X)^{-1}.$$
- Show how this formula simplifies under homoscedasticity. **[6 marks]**
- (c) Use part (b) to derive $\text{Var}(\hat{\theta} \mid X)$. Let $\widehat{\text{se}}_{\text{White}}(\hat{\theta})$ denote its estimated White standard error. Write down the corresponding test statistic for testing $H_0 : a'\beta = \theta_0$ and describe when we would reject the test. What pattern of heteroscedasticity do White standard errors allow for? Are they valid in the presence of autocorrelation? **[8 marks]**
- (d) Explain the standard trade-off between type I and type II errors. Why does lowering the significance level α make a type I error less likely but a type II error more likely? **[5 marks]**

Question 5

A university invites students to volunteer for a study of a new online revision programme, then selects 150 students at random from the population of volunteers. Within this group, 90 students are randomly assigned access to the programme ($D = 1$), while 60 are assigned to the control group ($D = 0$). Let Y be the end-of-term score. For the population of volunteers, consider the regression

$$Y = \alpha + \beta D + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D] = 0.$$

The realized mean score is 78 in the programme group and 71 in the control group.

[25 marks]

- (a) Use $\mathbb{E}[\varepsilon \mid D] = 0$ to show that
- $$\alpha = \mathbb{E}[Y \mid D = 0] \quad \text{and} \quad \beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$
- [5 marks]**
- (b) In the regression above, calculate $\hat{\alpha}$ and $\hat{\beta}$. Briefly explain what $\hat{\beta}$ compares. **[4 marks]**
- (c) For a student i , let $Y_i(1)$ denote what the score would have been if the student had received access to the programme, and let $Y_i(0)$ be the score if they had not received it.
- Write the observed outcome Y_i in terms of D_i , $Y_i(1)$ and $Y_i(0)$.
 - State the random-assignment condition using D_i , $Y_i(1)$ and $Y_i(0)$.
 - Use random sampling and random assignment to show that
- $$\beta = \mathbb{E}[Y(1) - Y(0)],$$
- where the expectation is for the population of volunteers. **[7 marks]**
- (d) Does random assignment make the 150 student volunteers representative of all students at the university? Explain the different roles of random assignment and random sampling. **[5 marks]**
- (e) Now suppose instead that the university takes a random sample of 150 students, but each student chooses whether to use the online revision programme. What would random sampling achieve? Why would the difference in mean scores not automatically have a causal interpretation? **[4 marks]**

Question 6

Let n be even. The independent observations $X_1, \dots, X_n$ all have mean μ and are Normal, but their variances alternate between two positive constants:

$$\text{Var}(X_i) = \begin{cases} v_L, & i \text{ odd}, \\ v_H, & i \text{ even}, \end{cases} \quad 0 < v_L < v_H.$$

Assume that v_L and v_H are known. For $w > 0$, consider the weighted mean

$$T_n(w) = \frac{\sum_{i \text{ odd}} X_i + w \sum_{i \text{ even}} X_i}{(n/2)(1+w)}.$$

Thus each group contains $n/2$ observations. The odd-indexed observations form the lower-variance group, while w is the weight on each observation in the higher-variance group relative to an odd-indexed observation. The denominator normalizes the weights so that they sum to one.

[25 marks]

- (a) For arbitrary $w > 0$, calculate $\mathbb{E}[T_n(w)]$ and show that $T_n(w)$ is an unbiased estimator of μ .

[4 marks]

- (b) For arbitrary $w > 0$, let

$$S_n(w) = \sum_{i \text{ odd}} X_i + w \sum_{i \text{ even}} X_i$$

denote the numerator of $T_n(w)$. Show that $\text{Var}[S_n(w)] = (n/2)(v_L + w^2 v_H)$. Hence derive the variance of $T_n(w)$. Explain why $T_n(w)$ is Normally distributed and state its exact distribution.

[6 marks]

- (c) Show that the value of w that minimizes the variance of $T_n(w)$ is $w^* = v_L/v_H$. You may use $\frac{d}{dw}\{(v_L + w^2 v_H)/(1+w)^2\} = 2(wv_H - v_L)/(1+w)^3$. Verify that this is a minimum, and give some intuition for why the higher-variance observations receive less weight.

[6 marks]

- (d) Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Notice that $\bar{X}_n = T_n(1)$. You may use $\text{Var}(\bar{X}_n) = (v_L + v_H)/(2n)$. Use parts (b) and (c) to show that $\text{Var}(\bar{X}_n) - \text{Var}[T_n(w^*)] = (v_H - v_L)^2/[2n(v_L + v_H)]$. Hence show that the optimally weighted estimator has strictly smaller variance under the maintained assumption $v_L < v_H$. If equal variances were allowed, when would the two estimators have the same variance? Give an intuitive explanation.

[5 marks]

- (e) For a fixed $w > 0$, use the mean and variance from parts (a) and (b) to show that $T_n(w) \xrightarrow{p} \mu$. Then state the exact distribution of $\sqrt{n}[T_n(w) - \mu]$, and hence its asymptotic variance. What is this asymptotic variance at $w = w^*$?

[4 marks]

END OF PAPER